

| LIBRARY MANAGEMENT SYSTEM |                                                                                                          |                                                                        |                                                                         |
|---------------------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Features                  | Comprehensive coverage                                                                                   | Risk-based approach                                                    | Continuous Improvement                                                  |
| User Login                | Test valid login, invalid login, empty fields, and session timeout                                       | Prioritize invalid login and security scenarios (high risk)            | Re-test after security updates or password policy changes               |
| Member Registration       | Test valid details, duplicate members, and missing data                                                  | Duplicate member prevention tested first (medium risk)                 | improve validation based on user errors                                 |
| Search book               | Test exact book name (finds it), half name (finds list), no book (shows empty), blank search (all books) | Test no book found first (low risk)                                    | Partial - Test quick search daily, full check monthly                   |
| Return Book               | Test book issued (returns ok), book overdue (fine added), book already returned (error)                  | Test overdue return first (fnormal risk)                               | Yes - Test return after issue or fine updates                           |
| Generate Report           | test overdue books report (shows list), no overdue (empty), export to file                               | Test overdue report first (admin issue)                                | No - Not Needed (run reports weekly, test before use)                   |
| User Register             | Test new user (adds), same email (error), bad email (error)                                              | Test same email first (data problem)                                   | Partial - Quick check daily                                             |
| Borrowing book            | Test normal borrowing, user with too many books, and expired memberships.                                | Prioritize testing the "Max book limit" (high risk of inventory loss). | Re-verify after any changes to borrowing or membership rules.           |
| Returning book            | Test returning on time, returning late (fines), and returning damaged books.                             | Prioritize "Fine calculation" logic (financial risk).                  | Monthly check to ensure returned books show as "Available" immediately. |
| Add Book                  | Test valid data, duplicate ISBN, missing fields, invalid category                                        | Test duplicate ISBN first (data integrity risk)                        | Yes – Verify add book after DB or schema changes                        |
| Delete Book               | Test delete available book, issued book (block), invalid book ID                                         | Test deleting issued book first (system consistency risk)              | Partial – Monthly full delete validation                                |
| Update Book               | Test update title, author, category, invalid values                                                      | Test ISBN update first (uniqueness risk)                               | Yes – After catalog structure changes                                   |
| Issue Book                | Issue available book, already issued book, invalid user, limit exceeded                                  | Test issuing already issued book first (core flow risk)                | Yes – Every build                                                       |